Ignytion io

IGNYTION IO · TECHNICAL LAB NOTE

I2C protocol: From RTL to routed GDS

A beginner's walkthrough of a checked open-source reference build

Follow i2c_core from its communication rules and Verilog **source** to a routed IHP SG13G2 layout. Worked calculations, measured digital traces and tool results are identified separately. This is an educational block, not a manufactured chip or a signoff-ready design.

STAGE	WHAT IT PRODUCED	RESULT
Simulation	Listed directed checks	PASS directed
Synthesis	1007 mapped functional cells	PASS mapping
Routing	Routed DEF and extracted SPEF	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	Decoded placement agreement	PASS export
Foundry DRC / LVS	No accepted result	UNPERFORMED

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What is in this document

Start with the wire-level rules, then follow the same circuit through the tools. Each section has a specific evidence boundary.

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SECTION 1 · START HERE

Start here

A protocol is an agreement about how connected devices exchange information: which wire carries each bit, when to sample it, and how to recognise a complete transfer.

WHAT WE BUILT AND WHAT IT PROVES

A Verilog description represents circuits, not software instructions running one after another. Registers remember values; combinational logic calculates the next values. A **clock** advances the registered state. The next pages first explain this protocol at its wires, then connect those rules to the actual reference RTL.

The implemented unit is i2c_core in hw/protocols/i2c. It is a fixed **reference** block, not the Jane Street programmable combined emulator or a Tiny Tapeout submission top. No new competition architecture, instruction set, pin map or authority decision is implied by these documents.

Arbitration is not decisively tested. Stretching and repeated **START** have indirect data/completion coverage only. Later pages give the exact stimulus and the implementation evidence rather than using a broad protocol-compliant label. A successful mapped netlist, routed DEF and decoded GDS establish concrete implementation/export results, not readiness to manufacture.

The reading path starts with chip-design vocabulary, then develops the protocol, calculations, local interface and failure cases. Separate sections explain lint, simulation, synthesis, placement, routing, timing and **GDS**. The last two pages define design and physical terms. No prior ASIC-flow experience is assumed.

EVIDENCE VOCABULARY

Label	Meaning in this document
Worked example	Explicit assumptions; independently calculated
Measured trace	Positive digital regression VCD, where present
Directed PASS	Listed assertions and negative controls
Routed / decoded GDS	Physical export and structural agreement
Signoff	Not performed or claimed

SECTION 2 · CHIP DESIGN

Chip design in ten minutes

Each stage makes the same circuit more concrete. The folders below are the actual unit layout, not a copied APB command list.

CHIP DESIGN

RTL states what hardware should do. A simulator checks selected input sequences before any silicon exists. Synthesis replaces that description with gates from a process library. Physical tools choose positions and metal routes. Timing analysis checks the resulting delays; stream-out writes the mask-layout geometry.

THE PIPELINE, STAGE BY STAGE

FOLDER	WHAT HAPPENS HERE
00_lint	Read code for risky patterns; retained warnings
01_rtl	Pinned source and the local wrapper
02_sim	Independent stimulus, comparisons and fatal controls
03_verification	Supplemental checks; not a universal UVM PASS
04_synthesis	Map functional logic to SG13G2 cells
05_dft	Manufacturing test work remains unperformed
06_floorplanning	Die outline, rows, power and pin locations
07_placement	Choose legal positions for mapped cells
09_place_and_route	Clock tree, routing and SPEF extraction
08_sta	Read post-route TT and slow timing reports
10_gds	Stream-out and independent placement decoder
11_physical_verification	Foundry DRC / LVS remain unperformed

SECTION 3 · ELECTRICAL RULES

Drive low, or release

Open-drain ownership is a wired-bus rule. The pad-level implementation remains outside this digital reference.

ELECTRICAL RULES

When any **open-drain** participant pulls the shared wire low, the connected receivers observe a low. This supports shared ownership without opposing high and low drivers. Pull-up selection depends on rise time, bus capacitance and sink current. An RTL tri-state model is ideal digital behaviour and does not prove these analogue limits. The integration needs correct output-enable polarity.

These are stated classroom assumptions, not measured board values. The 0.8473 factor describes the 30-to-70 percent RC rise interval. A real design must use the device data sheet limits and include wiring, connector and pin capacitance. NXP UM10204 section 7.1 describes this method. This calculation does not prove electrical compliance of the **reference** RTL pads.

The pull-up can raise the line only when neither participant sinks current. A digital simulation does not model pull-up resistance or pad sink current.

The pull-up bounds are Rmin = (VDD - VOL) / IOL and Rmax = tr / (0.8473 x Cbus). With the stated values these give 2.9 / 0.003 = 967 ohm and 300 ns / (0.8473 x 100 pF) = 3.54 kohm. Choosing 2.2 kohm predicts tr = 0.8473 x 2200 x 100 pF = 186 ns. Doubling the capacitance doubles that prediction to 373 ns and exceeds the assumed 300 ns budget.

WORKED PULL-UP BUDGET

Assumption / result	Value
VDD, VOL, sink capability	3.3 V, 0.4 V, 3 mA
Cbus, assumed rise-time limit	100 pF, 300 ns
Rmin / Rmax	967 ohm / 3.54 kohm
2.2 kohm rise time / current	186 ns / 1.32 mA

SECTION 4 · TRANSACTION RULES

Read a register, not just a byte

A device address selects the target; the application register selects the value within it.

TRANSACTION RULES

The register index and auto-increment behaviour belong to the target's application contract, not to I2C itself. The local target model stores bytes for this exercise. A real device datasheet determines whether this exact multi-byte sequence is valid.

The first acknowledged read requests more bytes. The controller should NACK the final byte before **STOP** to tell the target that the receive phase is over. The regression checks A5 and 5A. Later uninitialised locations print unknown values but are not asserted, so they cannot count as extra verified payload bytes.

Repeated **START** keeps the transaction active while changing direction. The target owns read data, and the controller owns the final response. Some devices use different register widths or pointer behaviour; this worked convention is not universal.

HYPOTHETICAL TARGET 0X50, REGISTER 0X01

Phase	Byte / bus owner
Address write	A0 from controller; ACK from target
Register selection	01 from controller; ACK from target
Repeated START, address read	A1 from controller; ACK from target
Final value	A5 from target; NACK from controller
Release bus	STOP from controller

SECTION 5 · TIMING CALCULATION

Nominal rate versus wire rate

The simulation and physical clocks differ. Divider arithmetic alone does not describe the measured bus period.

TIMING CALCULATION

The five-phase divider gives nominal SCL = fwb / [5 x (PRER + 1)]. At 50 MHz, PRER 99 yields 100 kHz and PRER 24 yields 400 kHz before other delays. These are worked configurations, not independently measured board rates. The regression instead runs at 100 MHz with PRER 0x00C8, or 200: 1005 core clocks predict 10.05 us and 99.502 kHz.

The positive VCD contains steady address-**clock** intervals of 10.71 us, or 1071 core cycles and 93.371 kHz. The observed difference is 66 clocks per period, approximately 6.57 percent extra period. Input filtering and synchronisation are possible contributors; the exact state-by-state attribution has not been established. No causal claim is made from the rate difference alone.

Nine clocks are consumed by eight data bits plus **ACK**. The nominal maximum payload-bit fraction is therefore 8/9, before address, START, STOP and idle overhead. A hypothetical single-byte register read uses four nine-clock byte phases, giving 360 us at an ideal 100 kHz, excluding boundary gaps and stretching.

KEEP THE TIMING CONTEXTS SEPARATE

Context	Clock / result
Physical reference constraint	50 MHz; 20 ns
Simulation source clock	100 MHz; always #5
Nominal simulation divider	10.05 us; 99.502 kHz
Measured steady SCL	10.71 us; 93.371 kHz

SECTION 6 · RTL INTERFACE

Registers, command bits and pads

The local parallel interface drives a protocol engine; it is not itself the I2C wire protocol.

RTL INTERFACE

i2c_core wraps i2c_master_top, followed by the byte and bit controllers. The wrapper adds synchronised **reset** release and preserves the existing port contract.

Each command field has a separate bit assignment. Bit seven is **START**, bit six STOP, bit five READ, bit four WRITE, and bit three the response selection. The current bench uses these constants; verify status TIP clears before issuing the next command. A polling loop also needs a system timeout, which is separate from the bare peripheral logic.

ARST_LVL is an elaboration parameter. The default is zero, so low arst_i asserts **reset**. The wrapper converts polarity and passes active-low reset to the upstream controller. A two-flop release improves clock-domain reset handling but is not a metastability or reset-domain signoff report. There is also a synchronous Wishbone reset input. Pad integration still needs an actual I/O cell and electrical design.

SOURCE / COMMANDS

```
wire sta = cr[7];
wire sto = cr[6];
wire rd = cr[5];
wire wr = cr[4];
wire ack = cr[3];
wire sl_cont = cr[1];
wire iack = cr[0];
```

hw/protocols/i2c/01_rtl/i2c_master_top.v:237

SECTION 7 · CONTROL SEQUENCE

Bit control has several phases

The source FSM and the bit divider explain when SDA may change and when the bus is sampled.

CONTROL SEQUENCE

The table summarises the actual named states and output-enable assignments, not a newly invented controller. An enable of one releases the **open-drain** line; it does not force a high voltage. In wr_c the implementation checks SDA for arbitration. Command acknowledgement marks completion of the bit operation, which is separate from target acknowledgement on the ninth bus pulse.

The Wishbone read returns the receive register at address three; writing there queues outgoing data. This shared address is not an error. The bus acknowledgement means the register operation completed, not that a remote I2C target acknowledged.

The XOR converts the selected external assertion level into an internal active-low **reset**. While asserted, both release flops clear. Once the external reset becomes inactive, ones shift through on consecutive clock edges. The controller receives the second stage. This reduces release hazards inside this clock domain; it does not provide board-level reset sequencing for every pad or external target.

SOURCE-DERIVED WRITE-BIT SEQUENCE

State	SCL action	SDA action
wr_a	Pull low	Set current data bit
wr_b	Release	Keep data, allow settling
wr_c	Remain released	Keep data, enable bus check
wr_d	Pull low	Keep data, acknowledge bit command
idle	Await next command	Hold current drive policy

SECTION 8 · FAILURE ANALYSIS

A NACK is an observation

An unsuccessful transaction must be diagnosed at the address, ownership and timing levels.

FAILURE ANALYSIS

The bench probes an invalid address and asserts the NACK result. That test is useful because a controller can generate perfectly shaped clocks while communicating with no device. The simulation watchdog is a test safety mechanism, not proof of a hardware bus-recovery implementation.

The actual bus can remain low because another participant holds it. The RTL samples the pad input. The inherited bench includes a stretching target, but its completion assertion does not certify arbitrary stretch durations or stuck-low recovery. This limitation belongs in an test report.

The selected OpenCores design contains arbitration logic. Presence of that logic is different from test coverage. The second controller in this regression remains idle, so arbitration under contention is not validated.

COVERAGE BOUNDARY

Mechanism	Evidence strength
A5 / 5A reads and invalid-address NACK	Directed decisive checks
Repeated START and clock stretching	Indirect completion / data checks
Two-controller arbitration	Not tested
Later uninitialised model bytes	Not data-verified

SECTION 9 · MEASURED TRACE

An address and its acknowledgement

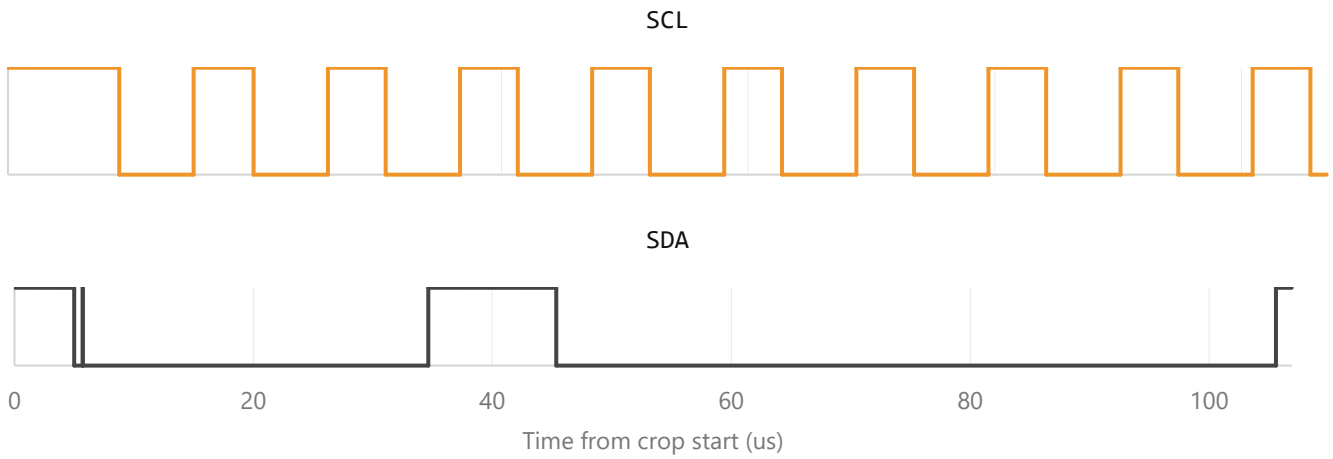
This chart is decoded from the positive digital regression, not an oscilloscope capture.

MEASURED TRACE

The selected transaction sends address byte 0x20. Splitting it into a seven-bit address and one direction bit yields target 0x10 and write direction zero. The decoder reads SDA at the first eight rising SCL edges, then separately checks the ninth sample. That response is low, establishing **ACK** for the plotted address phase.

The waveform is deliberately not relabelled as the hypothetical target 0x50 used in the register-read exercise. Its steady SCL interval is 10.71 us rather than the nominal 10.05 us calculated from the **prescaler**. Digital values in this trace cannot establish pull-up rise time, voltage thresholds, pad drive strength or electrical compliance. The source VCD hash is recorded in the document evidence manifest.

MEASURED POSITIVE REGRESSION



Logic HIGH=1, LOW=0. Digital simulation, not analogue compliance; source hash retained.

SECTION 10 · LINT

Lint: read code before running it

Verilator checks legal but suspicious code patterns without exercising transactions. A warning count is a review starting point, not proof of correctness.

LINT

The retained report contains 3 warning messages and 0 error messages. Categories present: CASEINCOMPLETE, SYNCASYNCNET, UNUSEDSIGNAL. These figures count report message headers; repeated warnings are not necessarily independent defects. The report is hashed in this document evidence manifest.

The invocation deliberately uses -Wno-fatal so warnings remain visible while lint execution can complete. A successful command therefore does not mean every warning was resolved. Width changes, incomplete cases, **reset** use and unconnected ports need review against the wrapper contract and directed tests.

Lint cannot show that a receive byte is correct, that a transaction finishes, or that an external electrical interface is safe. Those questions require a checking testbench, implementation evidence and separate board or signoff work.

RETAINED WARNING REVIEW

MEASURE	VALUE
Warning message headers	3
Error message headers	0
Clean warning-free RTL?	Not claimed
Protocol behaviour proved by lint?	No

SOURCE / COMMANDS

```
make lint
# inspect, do not just count
# 00_lint/lint_report.txt
```

00_lint/Makefile and run_flow.sh: retained Verilator report

SECTION 11 · SIMULATION AND VERIFICATION

What the I2C regression establishes

A testbench is code that drives the circuit and judges its outputs. These directed checks use independent expected data and deliberately failing controls.

SIMULATION AND VERIFICATION

The bench checks register values, successful A5 and 5A byte transfers, and NACK when the target model receives an invalid memory address. The existing second controller remains idle. The flow adds timeout and fatal comparisons, separate interrupt wires and integration through the real local wrapper.

Repeated **START** and forced clock stretching occur in the inherited sequence, but there is no independent edge-by-edge property proving every boundary or stretch duration. Third and fourth reads from uninitialised memory are not data-verified. These gaps remain visible rather than being folded into an exhaustive protocol claim.

Negative controls deliberately change expected data or a register result. Their purpose is to show that the checker rejects a wrong result. A passing negative-control harness means the deliberately corrupted child run failed as intended; it does not mean the corrupted DUT transaction was correct.

VERIFICATION EVIDENCE LOCATIONS

File / area	Interpretation
02_sim/test.v and positive log	Stimulus, expectations and actual PASS
02_sim negative-control logs	Expected fatal rejection
03_verification	Preserved source / supplemental checks
GATES.md	Acceptance records, not a replacement for logs

> **A checker must be able to fail**
>
> Completion alone is insufficient. Read the successful log and the expected fatal rejection together; neither is a protocol-compliance certificate.

SECTION 12 · SYNTHESIS

Synthesis: a description becomes gates

A netlist is a list of logic cells and their connections. Yosys translates the RTL into cells from the IHP SG13G2 library.

SYNTHESIS

A process design kit, or PDK, contains the process-specific files used by implementation tools. This **reference** uses IHP SG13G2, not the SKY130 library used by the golden APB primer. Standard cells are predesigned gates and storage elements with characterised timing and physical views.

The **source** is compiled and checked under the unit workspace before synthesis. Yosys maps the functional logic into the pinned IHP SG13G2 standard-cell library. The current netlist contains 1007 functional cells, excluding $scopeinfo metadata. The structural checker rejects remaining generic logic, latches or unresolved instances.

The mapped count excludes $scopeinfo metadata. It is not the count of every placed instance after **clock** buffers, repair cells and fillers are added. It also does not mean that the original protocol has been independently proven for all possible transactions.

MAPPING RESULT

MEASURE	CURRENT RESULT
Top module	i2c_core
Mapped functional cells	1007
Technology	IHP SG13G2
Unresolved generic logic / latches	Rejected by structural checker

SOURCE / COMMANDS

```
make synth
node verify_build.mjs synth
```

04_synthesis/synth.tcl; verify_build.mjs synthesis checks

SECTION 13 · PHYSICAL IMPLEMENTATION

Floorplan and placement

The floorplan gives the circuit a physical outline. Placement chooses where its cells sit inside that outline.

PHYSICAL IMPLEMENTATION

The die outline for this standalone block is 300 x 300 um. The floorplan establishes rows and **reference** power/pin geometry. The placer must fit cells legally without overlaps and with room for routing. This is a block-level educational layout, not a complete padframe or package design.

Mapped functional logic has 1007 cells. After physical additions and fillers, the routed DEF contains 14,546 placed standard-cell instances. A filler occupies empty row sites; it is not another functional logic gate. That distinction explains why the two counts differ so much.

Utilisation is the fraction of core area occupied by placed cell area. It is not simply functional-cell count divided by die size, and it changes between stages. This note does not substitute APB utilisation, row count or I/O count for unavailable protocol-specific measurements.

ACTUAL PHYSICAL REFERENCE

MEASURE	VALUE
Die outline	300 x 300 um
Mapped functional cells	1007
Placed standard-cell instances	14,546
Physical inputs	06_floorplanning; 07_placement
Placement agreement	DEF compared with decoded GDS

SECTION 14 · CLOCK AND WIRES

Clock tree and routing

Clock-tree synthesis distributes clocks to registers. Routing draws the metal connections between placed cells.

CLOCK AND WIRES

A register samples its input at a **clock** edge. Clock buffers distribute that edge across the block; the difference in arrival times is called skew. The reference is constrained at 50 MHz / 20 ns.

OpenROAD completes the staged physical flow. The **reference** die is 300 x 300 um. DEF records routed design geometry and cell placement; SPEF captures extracted parasitics used by the timing analysis. The scripts retain stage-specific output so the result can be investigated instead of relying on a final banner.

Real interconnect has resistance and capacitance, called parasitics. Extraction writes these in **SPEF** for post-route timing. The retained router-rule DRC file is empty. That checks the router view of its work; it is not a full foundry-deck check of the exported polygons.

READ THE PHYSICAL EVIDENCE

ARTIFACT	QUESTION IT ANSWERS
09_place_and_route/route.def	Where are cells and routed connections?
09_place_and_route/route.spef	Which parasitics were extracted?
Router-rule DRC report	Any reported routing-rule violations?
08_sta/timing_report.txt	What does extracted reference timing report?

> **Two checks can both be called DRC**
>
> The empty router report is accepted routing evidence. Foundry DRC is a separate unperformed obligation; one result cannot stand in for the other.

SECTION 15 · REFERENCE TIMING

Two measured reference corners

Static timing analysis adds cell and wire delays without simulating input patterns. Setup asks whether data arrives early enough; hold asks whether it stays stable long enough.

REFERENCE TIMING

The physical **clock** constraint is 50 MHz / 20 ns. Post-route timing uses the extracted design rather than an ideal pre-route wire estimate. The report at each corner must be inspected for worst setup and hold slack and for VIOLATED markers. The reported values are not a timing guarantee under unstated board, voltage or temperature conditions.

Current TT setup/hold slack is +15.81 / +0.17 ns. The slow-corner values are +14.55 / +0.21 ns. Both reports contain no VIOLATED marker. These numbers are taken directly from the local report files at document build time; the evidence manifest retains their hashes.

I/O delays are **reference** budgets, not a contract with an external target, transceiver or PHY. Bus synchronisation, reset recovery/removal, all additional corners and modes, signal integrity and board timing need their own analysis. Positive slack in these two reports is therefore called reference timing closure, not foundry or system signoff.

LIVE EXTRACTED SLACK, NS

Corner	Worst setup	Worst hold
TT	15.81	0.17
Slow	14.55	0.21
Status	PASS reference reports	No VIOLATED markers

> **Positive slack has a stated boundary**
>
> Slack is remaining timing margin. These values apply to the listed clocks, I/O budgets and two library corners; they are not a board-timing or manufacturing signoff guarantee.

SECTION 16 · GDS STREAM-OUT

Count the right objects

GDS stores layout polygons and hierarchical cell references. Stream-out combines routed geometry with the real standard-cell layouts from the PDK.

GDS STREAM-OUT

The **GDS** file contains 1,583,844 bytes and 48 structures. The independent decoder finds 14,546 placed standard-cell instances including fillers. That placement count is intentionally much larger than the 1007 mapped functional cells: filler instances occupy unused placement sites and structures are unique hierarchy definitions.

The decoder compares exported cell identities, coordinates and orientations against the routed DEF. A deliberately altered placement/export expectation fails, including under Python optimisation, so acceptance is not implemented with removable assert-only checks. Raw file SHA256 is a run-specific fingerprint because stream-out timestamps can change.

The OpenROAD router-rule DRC report is empty. It is not a foundry-deck DRC result. LVS, antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing signoff remain unperformed. No empty routing report is promoted into a DRC-clean tapeout claim.

LIVE EXPORT MEASUREMENTS

Quantity	Current value
Mapped functional cells	1007
GDS bytes	1,583,844
GDS structures	48
Placed standard-cell instances	14,546
Die	300 x 300 um

SECTION 17 · UNPERFORMED SIGNOFF

DRC and LVS: what is still missing

Design rule checking asks whether geometry follows the process rules. Layout versus schematic asks whether the extracted layout circuit matches the intended netlist.

UNPERFORMED SIGNOFF

Foundry DRC requires the applicable, qualified rule deck and an inspected report on the final polygons. Typical checks include minimum widths, spacing and enclosure. No foundry-deck DRC result has been accepted for this protocol **reference**. The router report is narrower evidence.

LVS requires extracting a circuit from the layout and comparing devices and connectivity with the intended circuit. A structural **GDS** decoder checking cell references, coordinates and orientations does not perform that extraction or connectivity comparison. No accepted LVS match is available.

Antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing release remain unperformed. The golden APB primer reports its own separate checks; those results do not transfer to SG13G2 or to this design.

SEPARATE ACCEPTED EXPORT FROM MISSING SIGNOFF

CHECK	THIS REFERENCE
Router-rule report	Empty; narrower routing evidence
Decoded GDS placement	PASS structural/export agreement
Foundry-deck DRC	UNPERFORMED
Extracted-layout LVS	UNPERFORMED
Manufacturing readiness	Not claimed

> **A GDS file is not tapeout approval**
>
> This package is suitable for studying and reproducing a block implementation. Do not hand it off as a signoff-clean manufacturing design.

SECTION 18 · RESULTS SUMMARY

Everything at a glance

Read each result together with the question it actually answers. The same PASS label never implies every protocol or manufacturing obligation is closed.

RESULTS SUMMARY

The directed functional result and the physical export result are both useful, but they prove different things. The physical build uses the fixed wrapper configuration and **reference** constraints. Additional modes, external peripherals, arbitrary clock relationships and board operation require their own evidence.

Important protocol boundary: Arbitration is not decisively tested. Stretching and repeated **START** have indirect data/completion coverage only.

FULL REFERENCE FLOW

STAGE	TOOL / EVIDENCE	STATUS
Lint	Verilator; 3 retained warnings	Review
Simulation	Independent directed expectations	PASS directed
Negative controls	Deliberate fatal mismatches	PASS rejection
Synthesis	Yosys; 1007 mapped cells	PASS mapping
Floorplan / placement	OpenROAD; 300 x 300 um	PASS reference
Routing / extraction	DEF and SPEF retained	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	KLayout; 1,583,844 bytes	PASS export
Foundry DRC / LVS	No accepted report	UNPERFORMED
Full-chip / board	No accepted integration proof	UNPERFORMED

SECTION 19 · RUN IT YOURSELF

Run it yourself in WSL

Use the pinned source, the unit tool lock and the external PDK revision together.

RUN IT YOURSELF

The proven entry point is a WSL login Bash shell. igny init prepares the packaging runtime. Unsetting CRUCIBLE_ENV after initialisation allows the unit crucible.toml workspace selection to control tool resolution rather than an inherited environment override. This is the observed workflow, not a change to Crucible internals.

Each unit has its existing environment and lock. The common tool set includes Icarus, Verilator, Yosys, OpenROAD and klayout-stream. The separate PDK checkout is IHP SG13G2 revision 22f43352dd8219f9007eb659e422e0d5fe28c5fb. A tool lock does not pin that external PDK by itself; IHP_PDK_ROOT must point to the matching installation.

The documented machine could not use the pip klayout package as the expected stream-out executable; igny tool install klayout-stream supplies the proven converter. make lint sim checks functional behaviour, make flow executes the staged implementation, and node verify_build.mjs all inspects artifacts. Review normal and intentionally failing logs before accepting the result.

Re-running a simulation can replace its VCD and invalidate an earlier document trace hash. Retain a document evidence manifest or regenerate the measured figures after the positive run. Do not silently preserve an old chart while claiming it is the newly generated waveform.

SOURCE / COMMANDS

```
igny init
unset CRUCIBLE_ENV
cd hw/protocols/i2c
igny workspace sync
igny tool install klayout-stream
make lint sim
make flow
node verify_build.mjs all
```

Run after entering the repository in WSL login Bash; PDK required.

SECTION 20 · SOURCES AND REMAINING WORK

Sources, ownership and remaining work

The pinned revision identifies the baseline; local integration changes are described separately.

SOURCES AND REMAINING WORK

Upstream: https://github.com/freecores/i2c
Pinned commit: 3b067f00ccced753b0502024766a51f58f3e04bc
Licence: Richard Herveille permission notice, 2001. Original **source** notices remain in the RTL. The document is authored by Ignytion; this does not transfer ownership of upstream code or remove its licence obligations.

Local modifications, directed test stimulus and **source** boundaries are recorded in hw/protocols/i2c/README.md and the repository append-only .context.md. The first three protocol lessons have separate editable 16:9 student decks; this A4 technical note is a reading and evidence document, not a replacement for their instructor notes.

Primary reading: NXP UM10204 Rev. 7 and TI SLVA689 for pull-up arithmetic.
Canonical reading: https://cache.nxp.com/docs/en/user-guide/UM10204.pdf and https://www.ti.com/lit/pdf/slva689

Open work remains explicit: Arbitration is not decisively tested. Stretching and repeated **START** have indirect data/completion coverage only. Full-chip integration, manufacturing checks and board validation are not closed by the standalone reference flow. The reader can reproduce the listed evidence without assuming those larger obligations have already been completed.

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SECTION 21 · DESIGN VOCABULARY

Glossary: design and protocol

Definitions for reading the protocol rules and the checked RTL. Physical implementation terms follow on the next page.

DESIGN AND VERIFICATION

TERM	WHAT IT MEANS
Protocol	Rules for exchanging information between devices
RTL	Registers and the logic moving values between them
Verilog	A language used to describe hardware
FSM	A circuit progressing through defined states
Lint	Static source checks; no transaction execution
Testbench	Code that supplies inputs and checks outputs
Coverage	Which defined situations were exercised
Negative control	An intentionally wrong case the checker must reject
VCD	Digital signal-change trace from a simulation
Handshake	Signals agreeing when data is accepted
ACK / NACK	Ninth-bit acceptance / non-acceptance response
PRER	Two-byte clock prescaler register
Open drain	Drive low or release; never actively drive high
Wishbone	Local register bus, distinct from the serial I2C bus

SECTION 22 · PHYSICAL VOCABULARY

Glossary: implementation and signoff

These definitions help distinguish a successfully exported reference layout from a manufacturing-ready design.

PHYSICAL IMPLEMENTATION

TERM	WHAT IT MEANS
Synthesis	Translate RTL into a technology-cell circuit
Netlist	Cells and the connections between them
Standard cell	Predesigned logic/storage cell in a process library
PDK	Process design kit: process-specific implementation files
SG13G2	IHP process used by these reference builds
Floorplan	Die/core outline, rows, power and pin geometry
Placement	Choose legal positions for cells
Filler	Physical row-site cell, not functional logic
CTS / skew	Clock distribution / difference in arrival times
Routing / via	Metal connections / links between metal layers
Parasitics / SPEF	Wire resistance/capacitance / extracted data format
Corner	A characterised process, voltage and temperature case
STA / slack	Static timing analysis / remaining timing margin
DEF / GDS	Placement/routing exchange / polygon hierarchy
DRC	Check geometry against applicable process rules
LVS	Compare the extracted layout circuit with its netlist
Signoff	Accepted applicable release checks; unperformed here

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